



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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Academic Year 2025 - 26

S8 PROJECT - II

FINAL Review

PROJECT TITLE

DIFFERENTIATED CASE MANAGEMENT

BIP PROJECT ID

2026IN285

CATEGORY

INTERNAL

BATCH MEMBERS

MANIKANDAN V (7376222CT129)

KUMAR V S (7376222CT127)

VISHALINI E (7376222CT162)

GUIDE

MURUGESAN P

Assistant Professor III

Department of Computer Technology

LITERATURE SURVEY

Sl.No.	References (Journal Papers Only)	Works carried out (with details of Methods/ Materials/ Software/ Algorithms / fabrication / techniques/ components used)	Information gathered relevant to your project
1	E-Governance Initiatives in Indian Judiciary (e-Courts Project).	- Digitization of court records - DCM-based software solutions.	Judicial Information Management Systems
2	Supreme Court of India, <i>Salem Advocate Bar Association v. Union of India</i> , Judgment on Case Management.	- Case Flow Management - Time-bound case disposal	Differentiated Case Flow Management
3	Law Commission of India, <i>Model Case Management Rules</i> , Jagannadha Rao Committee Report.	Categorization of cases	Rule-Based Case Categorization
4	R. Susskind, “The Future of Courts,” Oxford University Press, 2019	- Technology in modernizing judicial systems - Case prioritization and scheduling	Technology-Assisted Judicial Management

AIM & OBJECTIVES OF THE PROJECT

AIM:

To develop a software-based Differentiated Case Flow Management system that enables efficient case listing by prioritizing cases based on their characteristics rather than filing order.

OBJECTIVES:

1. To categorize cases based on complexity, urgency, and required adjudication time.
2. To provide multiple case processing tracks instead of a single uniform workflow.
3. To assist judges and court staff in efficient scheduling and prioritization of cases
4. To reduce delays and improve predictability in case disposal timelines
5. To support timely justice delivery by optimizing judicial resource utilization

ABSTRACT

The judicial system handles a wide variety of cases that differ significantly in complexity, urgency, and time required for adjudication. Treating all cases uniformly under a single case flow mechanism often leads to delays, inefficiencies, and case backlogs. The concept of Differentiated Case Flow Management (DCM) was introduced to address this issue by recognizing that not all cases require the same procedural treatment or time frame.

This project focuses on the development of a software system that supports Differentiated Case Flow Management for High Courts and Subordinate Courts. By categorizing cases based on predefined parameters such as case type, complexity, urgency, and statutory timelines, the proposed system aims to streamline case listing and scheduling. The software assists judges and court administrators in prioritizing cases efficiently, ensuring timely hearings and improving overall judicial productivity. The system is designed to reduce administrative bottlenecks and contribute to the speedy disposal of cases.

Problem Definition

Despite increased computerization in courts, the case scheduling and listing process has not seen significant improvement. Most courts continue to list cases primarily based on the chronological order of filing, without considering the nature or urgency of cases. This results in prolonged delays, inefficient use of judicial resources, and increased pendency.

Proposed Work and Methodology

Proposed Work and Methodology:

The project proposes the design and development of a web-based software system that implements Differentiated Case Flow Management. The system will:

- Classify cases into different categories (fast-track, normal-track, complex-track)
- Assign timelines and priorities based on case type
- Generate optimized case listing schedules
- Provide dashboards for judges and court administrators
- Maintain transparency and traceability of case progress

Proposed Work and Methodology

Methodology:

1. **Requirement Analysis**
Study existing court workflows and identify scheduling inefficiencies.
2. **System Design**
Design modules for case categorization, prioritization, scheduling, and reporting.
3. **Case Classification Logic**
Define rules based on case type, urgency, and statutory timelines.
4. **Software Development**
Implement the system using appropriate backend, frontend, and database technologies.
5. **Testing and Validation**
Test the system with sample case data to ensure accurate prioritization and scheduling.
6. **Evaluation**
Analyze improvements in case listing efficiency and predictability.

WORK CONTRIBUTION OF BATCH MEMBERS

- **Manikandan V (7376222CT129)**

Responsible for **requirement analysis and problem study**, including understanding the existing judicial case management process, identifying limitations in current case scheduling systems, and documenting functional requirements for the Differentiated Case Flow Management system.

- **Kumar V S (7376222CT127)**

Handled **system design and backend development**, including database design, case categorization logic, and implementation of differentiated case flow rules based on case type, priority, and timelines.

- **Vishalini E (7376222CT162)**

Worked on **frontend development and user interface design**, creating user-friendly dashboards for judges and court administrators, and contributed to testing, validation, and report preparation.

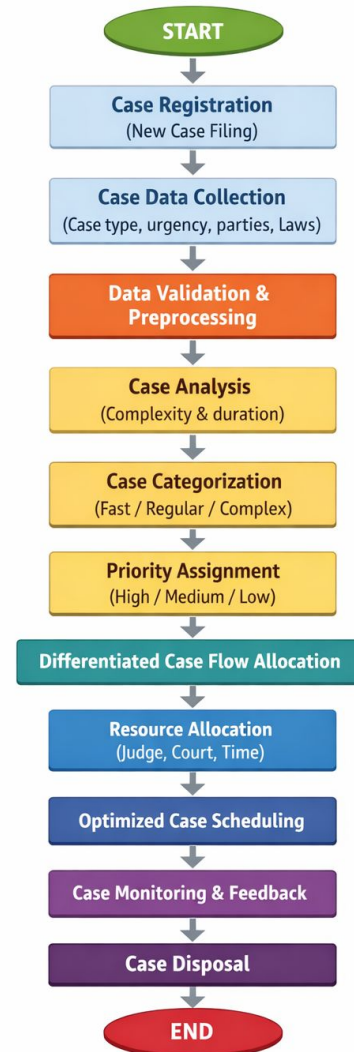
SCOPE OF THE PROJECT

- **Case Categorization:**
Automatically classify court cases based on complexity, urgency, nature of dispute, and estimated time required for disposal.
- **Optimized Case Scheduling:**
Enable efficient scheduling by allocating judicial time and resources according to case priority, reducing delays in simple and urgent cases.
- **Judicial Resource Utilization:**
Assist courts in better utilization of judges, courtrooms, and staff by balancing workloads and avoiding unnecessary congestion.
- **Decision Support for Courts:**
Provide analytical insights and recommendations to judges and administrators for case listing, monitoring, and performance evaluation.
- **Reduction of Case Backlog:**
Improve overall case disposal rates and ensure timely justice delivery by streamlining case flow and minimizing pendency.

NEED FOR THE CURRENT STUDY

- **Rising Case Backlog:**
Courts handle all cases through a uniform process, regardless of their complexity or urgency, leading to prolonged delays and accumulation of pending cases.
- **Inefficient Case Scheduling:**
Simple and time-bound cases often wait as long as complex matters, resulting in avoidable adjournments and underutilization of judicial time.
- **Lack of Decision-Support Systems:**
Current court management systems do not provide analytical support to help judges and administrators prioritize cases based on objective parameters.
- **Uneven Judicial Workload:**
Absence of workload balancing leads to overburdened judges in some courts while others remain underutilized.
- **Demand for Timely Justice:**
Delayed justice affects public trust in the legal system, making it essential to adopt intelligent, technology-driven solutions for efficient case flow management.

PROPOSED METHODOLOGY (Flow Chart)



PROPOSED METHODOLOGY (Gantt Chart)

	Week 1	WEEK 2	Week 3	Week 4	Week 5	Week 6	Week 7
Research	Compl... ▾	Compl... ▾					
Content		Compl... ▾	Compl... ▾				
Design			Compl... ▾	Compl... ▾			
Uploading				Compl... ▾	Compl... ▾		
Developing					Compl... ▾		
Finalizing						In Prog... ▾	In Prog... ▾
Marketing							Not Sta... ▾

METHODS/TECHNIQUES EQUIPMENT USED FOR PROJECT DEVELOPMENT

- **Differentiated Case Flow Management (DCFM) Approach**
Cases are classified into different tracks (Fast, Regular, Complex) based on complexity, urgency, and expected disposal time.
- **Rule-Based Classification Technique**
Predefined rules (case type, number of hearings, legal provisions, urgency level) are applied to categorize cases efficiently.
- **Priority-Based Scheduling Algorithm**
Cases are scheduled according to priority levels to ensure urgent and simple cases are disposed of quickly.
- **Data Preprocessing and Validation**
Cleaning, filtering, and validating case data to improve accuracy and consistency.
- **Decision Support System (DSS)**
Analytical modules provide recommendations for case listing, resource allocation, and monitoring.
- **Performance Monitoring & Feedback Mechanism**
Continuous tracking of case progress and adjustment of categorization rules to improve efficiency.

DESIGN(S) (SOFTWARE ARCHITECTURE)

- The system follows a **Layered / Multi-Tier Architecture** for better modularity and scalability.
- **Presentation Layer** provides user interfaces for judges, court staff, and administrators to file cases, view status, and schedules.
- **Application Layer** contains core business logic, including case analysis, categorization, priority assignment, and workflow control.
- **Decision Support Layer** assists in differentiated case flow management by recommending scheduling and resource allocation.
- **Data Access Layer** manages secure communication between the application and the database.
- **Database Layer** stores case records, user details, schedules, and audit logs.
- **Security Layer** ensures authentication, authorization, and role-based access control.
- The architecture supports **easy maintenance, future AI integration, and system scalability**.

ANALYSIS OF RESULTS AND DISCUSSION

Analysis of Results

The proposed system effectively categorized cases into different tracks based on predefined rules such as case type, urgency, and expected duration. This rule-based classification improved case scheduling and ensured that simple and urgent cases were processed without unnecessary delay. The system also helped in better utilization of judicial resources by balancing the workload among courts and judges. Overall, the results show improved efficiency when compared to the traditional uniform case handling approach.

Discussion

The results indicate that a structured, rule-based case flow management approach is sufficient to improve judicial efficiency without relying on artificial intelligence or complex algorithms. By focusing on clear classification rules and priority-based scheduling, the system achieves reliable and transparent results. While the current system performs effectively, future enhancements could include more refined rule sets, statistical analysis, or optional analytics modules to further optimize case flow and support large-scale court environments.

CONCLUSION

The proposed Differentiated Case Flow Management (DCFM) system provides an effective solution to the problem of case backlog in the judicial system. By categorizing cases based on complexity and urgency, the system ensures that simple and time-sensitive cases are handled promptly.

The rule-based approach improves case scheduling and judicial resource utilization without interfering with judicial decision-making. It helps in reducing delays, minimizing adjournments, and balancing the workload among judges.

Overall, the system enhances efficiency, transparency, and timely delivery of justice, making it a practical and reliable decision-support tool for modern court administration.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 1 : (MANIKANDAN V - 7376222CT129)

1. **Performed requirement analysis and problem study**
2. **Studied the existing judicial case management process**
3. **Identified limitations in current case scheduling systems**
4. **Documented functional requirements for the DCFM system**

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 2 : (KUMAR V S - 7376222CT127)

1. **Designed the overall system architecture**
2. **Developed backend modules of the system**
3. **Created database design for case management**
4. **Implemented case categorization and differentiated case flow rules**
5. **Handled logic based on case type, priority, and timelines**

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 3 : (VISHALINI E - 7376222CT162)

1. **Developed the frontend and user interface**
2. **Designed dashboards for judges and court administrators**
3. **Assisted in system testing and validation**
4. **Contributed to documentation and report preparation**

PENDING WORKS AND PLAN FOR COMPLETION

1. Implementation of detailed case reporting and analytics modules
2. Development of case monitoring and status tracking dashboard
3. Completion of role-based access control and security features
4. Integration testing with sample or real-time case data
5. Performance evaluation of the scheduling and categorization process

Status on Partial Completion and Submission of Project Report

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
1	Cover Page & Title Page (Both are in same format)	<u>COVER PAGE</u>
2	Bonafide Certificate	<u>BONAFIDE</u>
3	Declaration	<u>DECLARATION</u>
4	Acknowledgement	<u>ACKNOWLEDGEMENT</u>
5	Chapter I – Introduction	<u>INTRODUCTION</u>
6	Chapter 2 – Literature Survey	<u>SURVEY</u>

Submission of Project Report

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
7	Chapters 3 - Objectives and Methodology	METHODOLOGY
8	Chapters 4 - Proposed work modules (Chapters name can be based on the work)	Proposed work
9	Chapter 5 - Results and Discussion	Results and Discussion
10	Chapters 6 - Conclusions & Suggestions for Future work	Conclusions
11	References	References
12	Appendices	Appendices

REFERENCES

(Journal Papers/ Books / Website in IEEE Format)

Journal

1. R. Singh and P. Karthik, “Modern case flow management systems for reducing judicial delays,” *International Journal of Technology and Management Research*, vol. 8, no. 1, pp. 90–98, 2025.
2. H. Gupta and S. Verma, “Rule-based judicial systems for efficient case categorization,” *International Journal of Computer Applications*, vol. 15, no. 5, pp. 55–63, 2024.
3. D. Kumar and A. Sharma, “Optimizing judicial workload through digital case management systems,” *Journal of Emerging Technologies*, vol. 15, no. 2, pp. 78–86, 2024.
4. S. Iqbal and M. Rahman, “Structured case flow management for improved judicial performance,” *International Journal of Advanced Computer Science*, vol. 12, no. 4, pp. 130–138, 2024.
5. V. Reddy and K. Naidu, “Real-time case tracking systems for judicial transparency,” *International Journal of Scientific Research in Computer Science*, vol. 11, no. 2, pp. 44–52, 2023.
6. Y. Wang and L. Zhang, “Digital judiciary systems: Enhancing efficiency and reducing backlog,” *Journal of Information Technology and Law*, vol. 9, no. 1, pp. 25–34, 2023.